

		Humanities	STEM	Social Sciences	Other	Average
GPT-NeoX	20B	29.8	34.9	33.7	37.7	33.6
GPT-3	175B	40.8	36.7	50.4	48.8	43.9
Gopher	280B	56.2	47.4	71.9	66.1	60.0
Chinchilla	70B	63.6	54.9	79.3	73.9	67.5
PaLM	8B	25.6	23.8	24.1	27.8	25.4
	62B	59.5	41.9	62.7	55.8	53.7
	540B	77.0	55.6	81.0	69.6	69.3
LLaMA	7B	34.0	30.5	38.3	38.1	35.1
	13B	45.0	35.8	53.8	53.3	46.9
	33B	55.8	46.0	66.7	63.4	57.8
	65B	61.8	51.7	72.9	67.4	63.4

Table 9: **Massive Multitask Language Understanding (MMLU)**. Five-shot accuracy.

we observe a lot of variance in performance, that may indicate that this benchmark is not reliable. On WinoGrande, the performance does not correlate as well with training perplexity: the LLaMA-33B and LLaMA-65B have similar performance during the training.

4 Instruction Finetuning

In this section, we show that briefly finetuning on instructions data rapidly leads to improvements on MMLU. Although the non-finetuned version of LLaMA-65B is already able to follow basic instructions, we observe that a very small amount of finetuning improves the performance on MMLU, and further improves the ability of the model to follow instructions. Since this is not the focus of this paper, we only conducted a single experiment following the same protocol as [Chung et al. \(2022\)](#) to train an instruct model, LLaMA-I.

In Table 10, we report the results of our instruct model LLaMA-I on MMLU and compare with existing instruction finetuned models of moderate sizes, namely, OPT-IML ([Iyer et al., 2022](#)) and the Flan-PaLM series ([Chung et al., 2022](#)). All the reported numbers are from the corresponding papers. Despite the simplicity of the instruction finetuning approach used here, we reach 68.9% on MMLU. LLaMA-I (65B) outperforms on MMLU existing instruction finetuned models of moderate sizes, but are still far from the state-of-the-art, that is 77.4 for GPT code-davinci-002 on MMLU (numbers taken from [Iyer et al. \(2022\)](#)). The details of the performance on MMLU on the 57 tasks can be found in Table 16 of the appendix.

OPT	30B	26.1
GLM	120B	44.8
PaLM	62B	55.1
PaLM-cont	62B	62.8
Chinchilla	70B	67.5
LLaMA	65B	63.4
OPT-IML-Max	30B	43.2
Flan-T5-XXL	11B	55.1
Flan-PaLM	62B	59.6
Flan-PaLM-cont	62B	66.1
LLaMA-I	65B	68.9

Table 10: **Instruction finetuning – MMLU (5-shot)**. Comparison of models of moderate size with and without instruction finetuning on MMLU.

5 Bias, Toxicity and Misinformation

Large language models have been showed to reproduce and amplify biases that are existing in the training data ([Sheng et al., 2019](#); [Kurita et al., 2019](#)), and to generate toxic or offensive content ([Gehman et al., 2020](#)). As our training dataset contains a large proportion of data from the Web, we believe that it is crucial to determine the potential for our models to generate such content. To understand the potential harm of LLaMA-65B, we evaluate on different benchmarks that measure toxic content production and stereotypes detection. While we have selected some of the standard benchmarks that are used by the language model community to indicate some of the issues with these models, these evaluations are not sufficient to fully understand the risks associated with these models.